

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

Project Title – Research Agent

Domain of Project – Artificial Intelligence (AI) & Research Automation

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Problem Statement -

Problem Statement : Research Agent

The Challenge- A Research Agent is an AI system designed to assist with academic and scientific research tasks. It can autonomously search for literature, summarize papers, and organize references. Using natural language processing, it understands research questions and retrieves relevant information.

The agent can generate reports, suggest hypotheses, and even draft sections of research papers.

It saves time by automating repetitive tasks like citation management and data extraction.

Research Agents enhance efficiency, accuracy, and innovation in both academic and industrial R&D.

Objectives

- Develop an AI-powered Research Agent using IBM watsonx.ai and IBM Granite.
- Enable intelligent literature search using RAG and trusted academic sources.
- Automatically summarize research papers and extract key findings.
- Generate structured research reports and literature reviews.
- Organize references and support citation management.
- Answer research-related questions using evidence-based information.
- Reduce the time required for academic and scientific research.
- Improve research efficiency, accuracy, and productivity through AI automation.

Proposed Solution -

The proposed solution is an intelligent Research Agent built on IBM watsonx.ai using IBM Granite as the large language model and IBM watsonx Orchestrate (BOB) to manage the agent's workflow and decision-making. The system integrates Retrieval-Augmented Generation (RAG) to retrieve relevant information from trusted research papers, journals, PDFs, and academic databases before generating responses.

Users can interact with the agent using natural language to search research literature, summarize papers, compare studies, generate literature reviews, organize citations, and create research reports. The combination of IBM Granite's language capabilities and RAG's document retrieval ensures responses are accurate, context-aware, and grounded in reliable sources. By automating repetitive research tasks, the Research Agent enhances productivity and supports better academic and industrial research.

Technology Used

- **IBM Granite model** -for eg–ibm-granite-3-2-8b –For natural language understanding, reasoning, and personalization
- **IBM Cloud** –It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment**for deploying the Nutrition Agent.
- **RAG**–RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** -For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) –For RAG-based nutritional data retrieval

Orchestrate componentUsed -

- **IBM watsonx Orchestrate** – Creates and manages the AI Research Agent, coordinates workflows, and handles user interactions.
- **IBM Granite LLM** – Understands user queries, generates summaries, explanations, reports, and evidence-based responses.
- **Retrieval-Augmented Generation (RAG)** – Retrieves relevant information from research papers, PDFs, journals, and trusted knowledge sources before generating responses.
- **Knowledge Base** – Stores research papers, academic documents, and technical resources for intelligent retrieval.
- **IBM Cloud Lite** – Hosts the AI services and provides the cloud infrastructure for development and deployment.

Agent Instructions Screenshot-

IBM watsonx Orchestrate

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AA

Home / Manage agents / Research Agent

Test agentliveDeploy

Research Agent

Model : GPT-OSS 120B – Open...

ProfileKnowledgeToolsetBehaviorChannels Preview

Notifications: 1 Dismissed | 1 total

Legal Notice +

View all

Profile

Describe what your agent does so people and other agents know when to use it.

Description*

An AI-powered Research Agent that assists users in academic and scientific research by searching trusted literature, summarizing research papers, organizing references, extracting key information, generating reports, and providing evidence-based insights using IBM Granite and Retrieval-Augmented Generation (RAG).

Build with ...Talk to ag...

Running in debug mode

3:30 PM

Hello, welcome to Research Agent !!

Accuracy of generated answers may vary. Please double-check responses.

Generate a literature review.

Organize citations in APA format.

1/3

Type something...

Knowledge

IBM watsonx Orchestrate

English

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live

Deploy

Research Agent

AI

Model : GPT-OSS 120B – Open...

Build with ...

Talk to ag...

AI

Profile

Knowledge

Toolset

Behavior

Channels Preview

Knowledge

Provide the specific knowledge your agent needs to accomplish its goal.

Knowledge sources

Add an existing knowledge source, upload files, or connect to content repositories to add knowledge to your agent.

Replace source +

Research Methodology

by IBMid-691001Q5L8

Helps the agent answer questions about research methods, hypothesis formulation, and study design.

Used by 1 agent

4:49 PM

Hello, welcome to Research Agent !!

Accuracy of generated answers may vary. Please double-check responses.

Generate a literature review.

Organize citations in APA format.

< 1/3 >

Type something...

Documentation

[Home](#) / [Manage agents](#) / [Research Agent](#) / Research Methodology

Research Metho...

Delete

Files

Description

Helps the agent answer questions about research methods, hypothesis formulation, and study design.

Detailed descriptions will help guide your agents to effectively use your knowledge.

Edit knowledge settings

[Learn more about best knowledge practices](#)

Documents (3)

			Add documents
File name	URL source	Date uploaded	
IEEEpaper.pdf		12/07/2026	
research methodology methods and techniq...		04/07/2026	
student-paper-setup-guide.pdf		04/07/2026	

Behaviour Guidelines :

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Research Agent

Model : GPT-OSS 120B – Open...

Profile

Knowledge

Toolset

Behavior

Channels Preview

Guidelines

Add rules that guide an agent's actions and responses.

Add Guideline +

Literature Search

Paper Summarization

Evidence-Based Answering

Citation & Reference Management

Research Report Generation

Chat with documents

Chat with documents directly in chat.

Build with ...Talk to ag...

Running in debug mode

3:38 PM

Hello, welcome to Research Agent !!

Accuracy of generated answers may vary. Please double-check responses.

Generate a literature review.

Organize citations in APA format.

Type something...

Agentic Workflow :

IBM watsonx Orchestrate

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Tools

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Agentic workflow 1

OverviewParameters

Name

Literature Search

Description

Search trusted research sources and return relevant papers.

A description enhances the effectiveness of your agent when it is deciding which tools to use.

What makes a good description?

Tool descriptions explain what each tool does and how to use the tool, but they don't affect how the tool functions. To help AI agents effectively choose the right tool to assist users, follow these guidelines when writing descriptions:

CancelDone

Tools to Research Agent :

IBM watsonx Orchestrate

English Contact Sales Buy Now 19 trial days left

Add tools to Research Agent

Overview Configuration details

Description

The Mary Technology MCP Server connects enterprise legal teams to structured legal intelligence extracted from litigation documents. Mary processes legal documents, including PDFs, emails, and compound document bundles, and extracts structured facts with full source provenance, linking every fact back to its exact document, page, and paragraph. Through this server, users can search and browse their legal matters, access auto-generated matter summaries including case overviews, key issues, people and parties, and additional notes, and ask Mary Assistant for drafting help and question answering by searching through matter facts. Mary supports key enterprise practice areas including personal injury, general litigation, employment, commercial, insurance, and more. Features:

- ask_mary
- view_matter_summary
- search_matters

Published
By: DEA Technology Pty Ltd

Categories
Legal Productivity Research

Languages supported
English

Resources
[Support](#) [Terms and Conditions](#)

Cancel Add to agent

Deployment :

IBM watsonx Orchestrate

English

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Pre-deployment summary

Review agent before available to the users. The deployed agent will take a snapshot of its tools and collaborator agents. To update them in the live agent, deploy it again.

Research Agent

Profile

Knowledge

Tools (1)

Collaborators (1)

Behaviour

Close

X

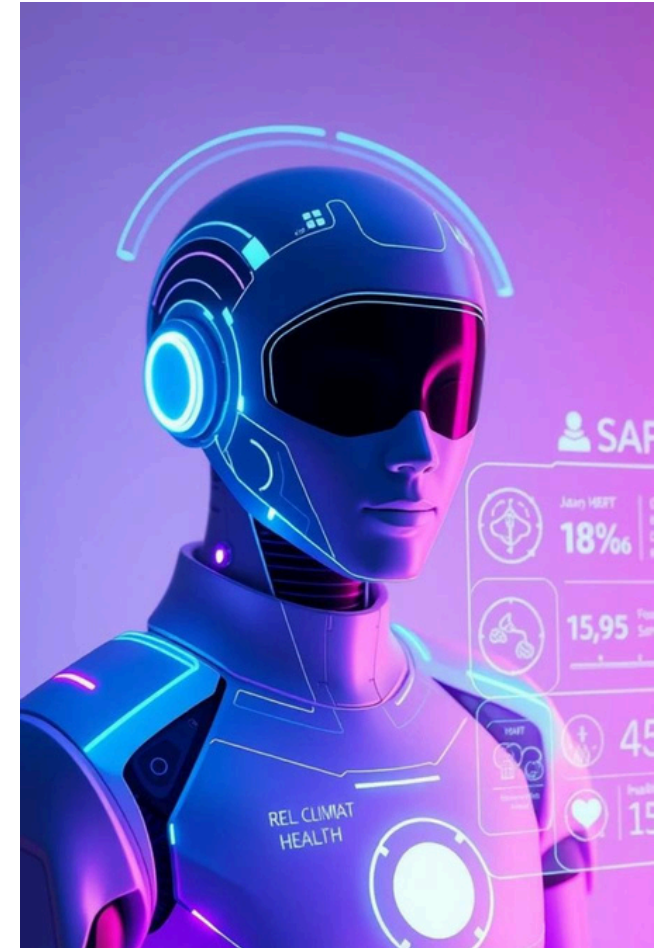
Cancel

Deploy

Role of Agentic AI in the solution

The Research Agent uses Agentic AI to understand user intent, decide the appropriate research task, retrieve relevant information from trusted sources, analyze research documents, and generate accurate, structured responses.

Instead of acting as a simple chatbot, the agent performs multi-step reasoning by searching knowledge sources, summarizing documents, organizing references, and producing research reports. This autonomous workflow reduces manual effort and enables faster, evidence-based research assistance.



Novelty and Uniqueness

- Integrates IBM watsonx Orchestrate, IBM Granite, and RAG into a unified AI research assistant.
- Retrieves information from trusted research documents instead of relying only on pretrained knowledge.
- Automates multiple research tasks, including literature search, paper summarization, citation management, and report generation.
- Provides conversational, evidence-based responses with structured outputs suitable for academic and professional use.
- Reduces the time and effort required to analyze large volumes of research literature while improving accuracy and productivity.

Future Scope

- Integrate with academic databases such as IEEE Xplore, Springer, PubMed, OpenAlex, and Crossref.
- Support multilingual research and automatic translation of research papers.
- Add AI-powered plagiarism detection and citation validation.
- Enable collaborative research workspaces with shared notes and document annotations.
- Generate presentations, posters, and visual summaries directly from research papers.
- Recommend research topics, identify research gaps, and suggest potential journals for publication.
- Integrate with reference managers such as Zotero, Mendeley, and EndNote.
- Develop a mobile application for researchers to access AI-powered research assistance anywhere.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Aniket Avdhut

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Jun 24, 2026
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Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Aniket Avdhut

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 24 Jun 2026 (GMT)

Learning hours: 30 mins

Certificates Earned:



Thank You!

Thank you for your time and interest.